Clinical Decision Support System for Disease Risk Assessment

Progress Report - 1

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Course: **CSIS 4495 – Applied Research Project**

Section: **03**

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1. Work Date / Hours Log

Date	Hours	Description of Work Done
Jan 27, 2026	3.0	Reviewed Week 4–5 milestones; researched on PDF extraction, Tesseract OCR, and Streamlit prior to implementation.
Jan 30, 2026	2.5	Designed preprocessing strategy for OCR-extracted medical text; identified relevant indicators and normalization needs.
Jan 31, 2026	2.5	Cleaned and structured extracted lab values; defined schema for post-extraction indicators Coordinated with Akiful to integrate preprocessing logic into the extraction pipeline and tested preprocessing against extracted outputs
Feb 2, 2026	3.0	Developed UI-only Streamlit prototype after extraction (layout, inputs, placeholders, disclaimer)
Feb 5, 2026	2.0	Disease research: heart disease and breast cancer indicators and their presence in patient-accessible reports Research done to find out any previously existing ML models based on the above 2 diseases
Feb 8, 2026	1.5	Reviewed Akiful's feedback and incorporated his contributions into the Streamlit interface to align UI with extraction workflow

2. Summary Description of Work Done during this Reporting Period

In Week 4, I went through the Week 4-5 milestones and studied PDF extraction, Tesseract OCR, and Streamlit. I developed and executed a preprocessing plan of OCR-extracted medical text, purified and formatted extracted laboratory values, and a schema of pertinent indicators. I worked with Akiful to incorporate the preprocessing logic into the extraction pipeline and tested it with extracted outputs. A problem with this step was that the extracted text had too much and unrelated information; this was overcome by using manual filtering and schema constraints that only allowed meaningful values to be retained. The approved project proposal was not adjusted.

In Week 5, I created a UI-only Streamlit prototype following the extraction phase, such as layout design, input fields, placeholders, and ethical disclaimers. I researched the information about the heart disease and breast cancer indicators and examined the already existing machine-learning models that relate to these diseases to guide future modelling. I also analysed the feedback of Akiful and added his input to the Streamlit interface, making the UI consistent with the extraction process. Problems were faced with Streamlit path and environment set-up and

problems with the installation of pandas, which were addressed by further trial and error, and functionality reliant on pandas was postponed preventing stalling. The validation of work was based on a single sample medical report, and further steps were aimed at the validation of multiple reports and the initiation of machine-learning model development and testing.

3. Repo Check-In of Implementation Completed

In this reporting period, I uploaded my implementation of Streamlit UI to the GitHub repository. They are the UI-only prototype that was created following the extraction phase and comprised of report upload elements, structured input fields, placeholder results, ethical non-diagnostic disclaimers, and documentation. Moreover, I created the preprocessing logic of OCR-extracted medical data and provided Akiful with this preprocessing logic so that he could use it in the extraction pipeline. I thus contributed the repository the ownership of the patient-facing interface and preprocessing design, and the code related to extraction was uploaded by my teammate independently.

The files/folders I have checked in the repo are as follows:

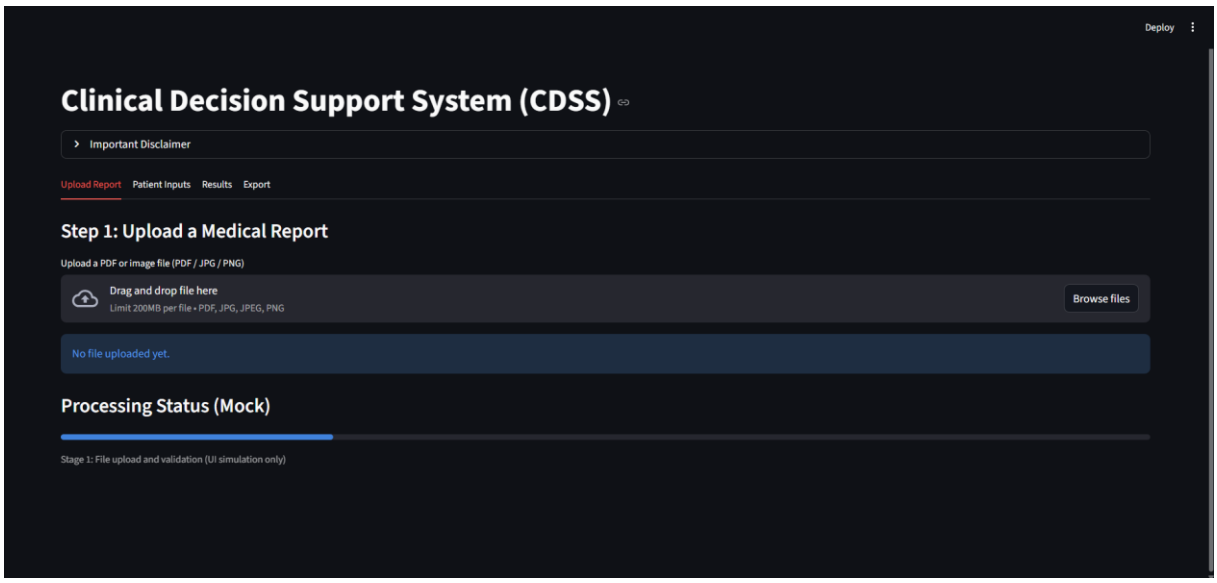
1. Implementation/sampleapp.py

4. AI Use Section

AI Tool	Version / Account	Purpose of Use	Value Added by Student
ChatGPT	GPT-5.2 (Premium)	Guidance on cleaning messy OCR-extracted medical report text and keeping only important information	Applied the guidance to design preprocessing logic, manually selected relevant indicators, and coordinated integration with the extraction pipeline
ChatGPT	GPT-5.2 (Premium)	Pointers on designing a simple Streamlit interface before backend	Independently designed and implemented the UI-only Streamlit prototype
ChatGPT	GPT-5.2 (Premium)	Understanding commonly reported clinical indicators for heart disease and breast cancer in patient-accessible records	Used the information to guide indicator selection for preprocessing and future machine-learning feature design

5. APPENDIX

1. My medical report text looks messy after OCR. Guide me on how can I clean it and keep only the important info
2. How should I design a simple Streamlit interface first, before connecting it to the backend, give me pointers?

3. What clinical indicators are commonly reported in patient-accessible medical records for assessing heart disease and breast cancer risk

UI screenshot